

Does AI really improve jet-engine health monitoring?

We built a head-to-head test with ground truth. Here is where AI wins, where it cannot — and what the difference is worth.

*One-page summary of **EHMbrAI**n: AI-based vs. traditional engine health monitoring, a pre-registered gas-path-analysis tested on the CFM56-7B with synthetic ground truth*

The question, made falsifiable

Does AI beat the traditional engine-health-monitoring stack — or is that marketing? We built the comparison no field study can run: a pyCycle thermodynamic twin of the CFM56-7B26 calibrated to certificate data, a 100-engine synthetic fleet (SynCFM56) with known ground-truth health, and two *tuned* practitioners — the classical stack (Kalman GPA, CUSUM, trending) and an AI suite (Mahalanobis, GRU, conformal) — given identical data, identical sensors, and a symmetric 50-trial tuning budget. Every hypothesis, threshold, and test was frozen in twelve tagged pre-registrations *before* the test set was touched; failed gates were repaired physically, never by moving thresholds.

Verdicts (single confirmatory pass, Holm-corrected)

Task	Trad.	AI	Verdict
Detection recall (FA 1=1)	0.13	0.48	AI, $p=.008$
median delay (cycles)	6033	499	12× earlier
Isolation, confusable faults	31 %	31 %	<i>tie: wall</i>
RUL RMSE @90 % life (cy)	1981	858	AI, $\delta=0.89$
vs. advanced classical	1118	858	AI, $p=.002$
same, @70 % life	1064	1059	<i>tie: $p=.41$</i>
90 % interval, both conformal (cy)	2717	2028	1.34× tighter
Physics-informed hybrid	—	—	refuted 2×

The pattern survives contact with reality: on NASA’s C-MAPSS benchmark the same pipelines score 14.9 vs. 42.1 cycles RMSE, and the classical side was even upgraded mid-study to a similarity-based prognostic (1981→1118) — AI still wins (858, $p=0.002$). Against that upgraded baseline the win is *late-life only* (a tie at 70 % of life), and the interval advantage is 1.34×, not 5.9×, once the classical side is given the same conformal calibration — a wrapper any method can adopt. Both corrections are the project’s own, found by re-auditing its headline numbers.

The split is physics, not fashion

AI wins exactly where the task is *statistical aggregation over time*: pulling faint trends out of noise (detection), integrating a whole degradation history into a forecast (prognosis), and honestly sizing its own error (uncertainty). It ties exactly where the *sensor set* cannot distinguish two faults: the cockpit suite’s influence-matrix geometry leaves seven con-

fusable fault pairs at $\sim 1.3^\circ$ separation, and no model — classical or learned — can invert information that never reached the data. Proof: adding two real station sensors lifts confusable isolation from 0.15 to 0.92 ($p=0.001$); adding *virtual* (model-derived) versions of the same sensors changes nothing ($p=1.0$). Where learning does add isolation power is fusion: a GRU fusing multi-flight physics projections more than triples classical multi-point analysis (0.65 vs. 0.17) on identical windows, and the ACARS reporting *schedule* itself becomes a design variable (+79 % separability from one extra stabilized-cruise report).

Certificates, not promises

Two by-products may outlast the scoreboard. An *identifiability certificate* — accumulated Fisher information over an engine’s actual flight history — predicts per-direction diagnosis error *before* deployment, validated against ground truth (Spearman $\rho=0.70$, though unprovable as tested: ≈ 0.24 of it is the matrix the bound shares with the estimator, ≈ 0.46 survives without it, and ten directions cannot reach significance): it tells an operator which faults their fleet can see and which sensor rescues which blind spot. A *prognostic floor* — the aleatoric RUL uncertainty that no model can remove — shows 87 % of early-life error is irreducible, while at 90 % of life the best method still sits 4× above the floor: buy algorithms for the endgame, calibration for the rest.

What it is worth

The operational currency is converting unscheduled removals into scheduled ones. Against the floor-derived ceiling, the AI stack converts a net 35–50 % of unscheduled events across realistic lead-time demands; the traditional stack manages 0–25 %. For a 50-aircraft, 100-engine fleet, a Monte-Carlo cost model prices the difference at \$3.2M/yr median (\$0.9M–\$8.0M p10–p90, <1 % probability of net loss).

Takeaway: “AI vs. traditional” is the wrong question. AI is a strictly better *statistician* on the same data; it is not a new *sensor*. Buy AI for detection, prognosis, and calibrated uncertainty; buy instrumentation for isolation; and demand certificates — identifiability and floor — that say, in advance, which regime you are in.

Full study: 106-page report, 12 pre-registration tags, every figure and number auto-generated from versioned artifacts, replicable end-to-end on a desktop (**make all**, ~ 11 min). Repository: github.com/cedarcreeks/EHMbrAI.